Krish Maniyar

+91-94290-48002 | krishmaniyar27@gmail.com | linkedin.com/in/krish-maniyar | github.com/krishmaniyar | krishmaniyar.vercel.app

Education

Vellore Institute of Technology (VIT) — Vellore, India
Bachelor of Technology in Computer Science and Engineering — *Aug 2023 – May 2027*

- CGPA: 9.27 / 10.0
- Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming, Software Engineering

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Dart
Frameworks & Libraries: FastAPI, React, Next.js, Flask, Flutter, LangChain, PyTorch, Scikit-learn, SQLAlchemy, Celery
Databases: PostgreSQL (pgvector), MySQL, MongoDB, Redis, ChromaDB, Firebase, Supabase
Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Git, Linux, Vercel, Prometheus
AI / ML: Hugging Face Transformers, Sentence Transformers, RAG, NLP, Vector Embeddings, HDBSCAN

Projects

Strategic News Analyzer | *FastAPI, Next.js, PostgreSQL, pgvector, Celery, Redis, Docker, Groq LLM* | GitHub

- Constructed an end-to-end geopolitical intelligence platform ingesting news from 5+ sources every 15 minutes via SHA-256 deduplication and a Celery Beat async pipeline, reducing duplicate storage overhead to near-zero.
- Orchestrated parallel AI analysis via `asyncio.gather()` – sentiment, bias, summarization, and 768-dim embedding generation simultaneously – storing results in pgvector (HNSW indexing) for sub-millisecond ANN retrieval.
- Architected a hybrid RAG engine fusing PostgreSQL vector ANN + full-text search via Reciprocal Rank Fusion (RRF), streaming token-by-token SSE responses to a Next.js dashboard; deployed on AWS EC2 + Docker + Vercel.

Query Weaver | *FastAPI, React, MySQL, ChromaDB, Sentence Transformers, Groq LLM* | GitHub

- Engineered a full-stack NL-to-SQL assistant converting plain-English questions to MySQL-compatible SQL; implemented vector-based schema retrieval (top-3 tables via ChromaDB semantic similarity) achieving 0.3–1.3s total pipeline latency.
- Designed a conversational memory system persisting Q&A history in ChromaDB with semantic retrieval, enabling multi-turn query refinement and reducing context hallucinations across sessions.
- Enforced SQL safety controls detecting destructive operations (INSERT/UPDATE/DELETE/DDL) with an explicit user-confirmation gate preventing accidental data loss; auto-synced schema embeddings after each DDL via SSE broadcast.

SurgWound Assist | *PyTorch, BiomedCLIP, PubMedBERT, Scikit-learn, FastAPI, ONNX* | GitHub

- Architected an 8-attribute multi-task clinical vision system using a dual-stream BiomedCLIP ViT-B/16 backbone fusing global and high-resolution wound-bed crops, achieving 54.29% Mean Macro-F1 and **100% Emergency Triage Recall** – surpassing GPT-4o zero-shot by 27.9%.
- Resolved severe 63:1 long-tail class imbalance via Fisher-consistent Logit-Adjusted Loss combined with a WeightedRandomSampler, eliminating majority-class collapse and enabling end-to-end anatomical location classification at 63.84% Macro-F1.
- Delivered inference in **1.8s on a single consumer GPU (<4.2 GB VRAM)** via ONNX Runtime, reducing hardware requirements by over 95% vs. the 7-MLLM (>80 GB VRAM) competing baseline.

Experience

Flutter Application Developer Intern — May 2025 – Jun 2025
Marwadi Shares & Finance Ltd. (MSFL) — *Rajkot, India*

- Developed the full Flutter mobile frontend for MChat, a real-time organizational chat application serving 150+ employees, integrating REST APIs and a .NET SignalR hub for WebSocket-based real-time messaging.
- Resolved SignalR transport failures in the UAT environment – diagnosed WebSocket incompatibility and implemented Long Polling as a fallback, restoring stable real-time connectivity and eliminating persistent connection drops.
- Implemented a heartbeat/keep-alive mechanism and lifecycle-aware reconnection strategy handling mobile app pause/resume transitions, ensuring uninterrupted message delivery across network interruptions.

Core Committee Member — Hackathon Management — Apr 2024 – Feb 2025
Centre for Social Entrepreneurship and Development (CSED), VIT — *Vellore, India*

- Co-organized a national-level hackathon with 200+ participants, driving a 60% boost in event reach through cross-functional coordination of logistics, sponsor outreach, and participant communication pipelines.

Achievements and Certifications

Hackathons: YANTRA Hackathon – Finalist — EcoMind (AI waste-management, Flutter + TFLite + Firebase + Gemini API) — 2025–26; ET AI Hackathon 2.0 – Participant
Certifications: Oracle OCI 2025 Foundations Associate; Oracle OCI 2025 AI Foundations Associate (Jan 2026); AI A-Z 2026: Agentic AI, Gen AI & RL – Udemy (Jan 2026); Data Science, ML, DL & NLP Bootcamp – Udemy (Oct 2025)